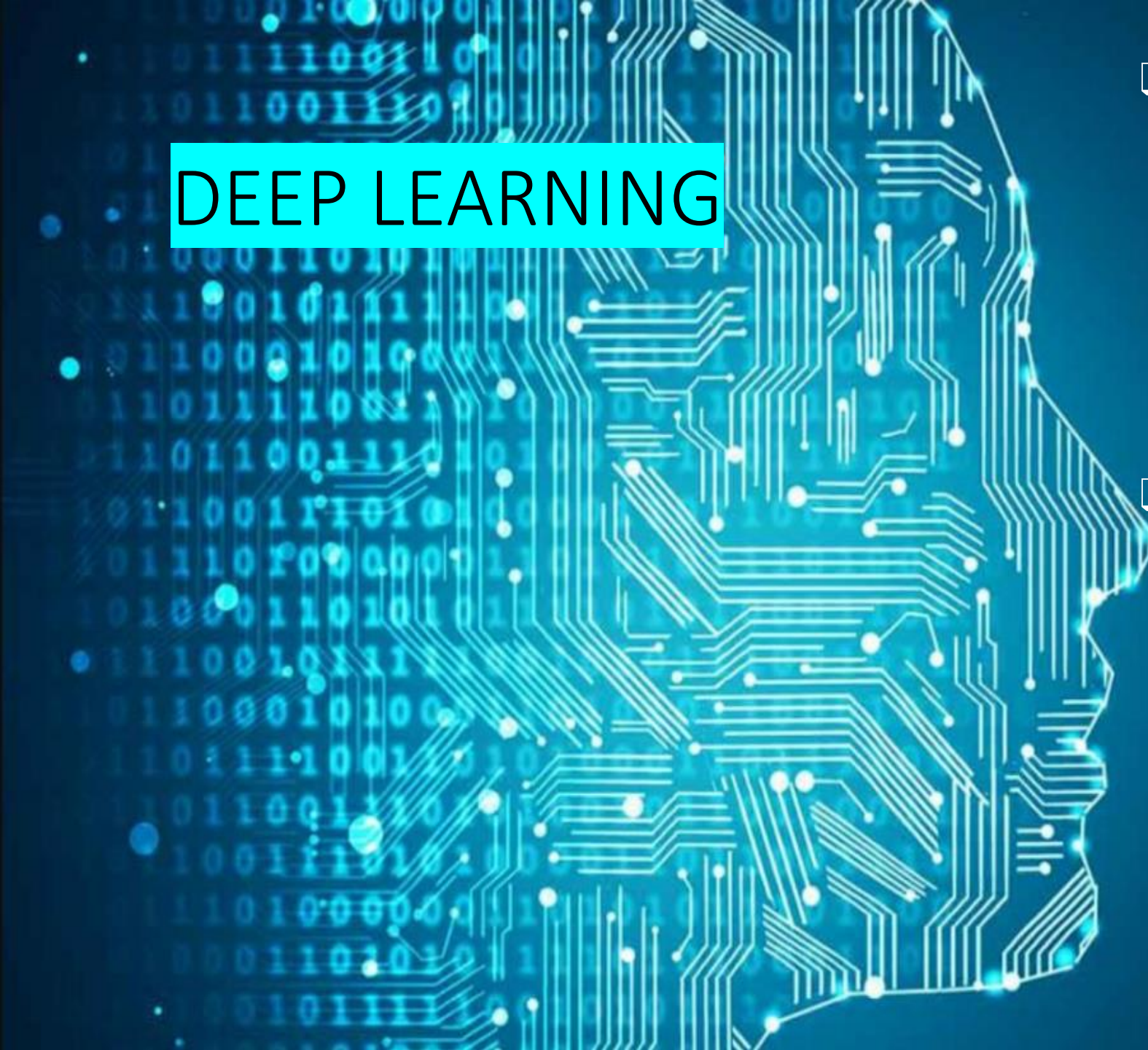




DEEP LEARNING

- ❑ Deep learning is a branch of artificial intelligence concerned with solving highly complex problems by emulating the working of the human brain.
- ❑ In deep learning, we use neural networks which use multiple operators placed in nodes to help break down the problem into smaller parts, which are each solved individually.

AI

Can a computer think? Then AI term was coined. Predefined rules back then. Tons of diff rules created for the AI.

Neural Network: A form of machine learning that uses a layered representation of data.

Data is transformed between diff layers. At different layers, different features will be extracted.

WHAT IS A NEURON

- ❑ The neuron is nothing more than a set of inputs, a set of weights, and an activation function.
- ❑ The neuron translates these inputs into a single output, which can then be picked up as input for another layer of neurons later on.
- ❑ Logistic regression is similar to a non-linear perceptron or a neural network without hidden layers.

NEURON – THE BUILDING BLOCK

- Each input is multiplied by a weight

$$x_1 \rightarrow x_1 * w_1$$

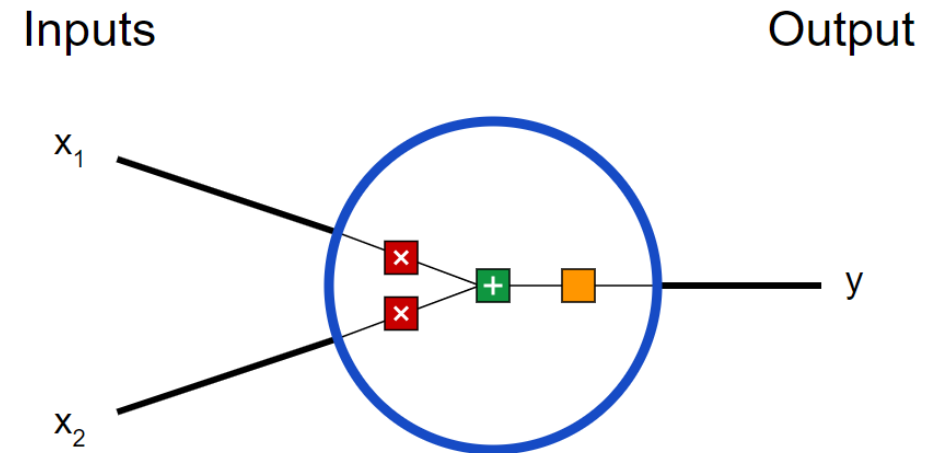
$$x_2 \rightarrow x_2 * w_2$$

- All the weighted inputs are added together with a bias b

$$(x_1 * w_1) + (x_2 * w_2) + b$$

- the sum is passed through an **activation function**

$$y = f(x_1 * w_1 + x_2 * w_2 + b)$$



ACTIVATION FUNCTIONS

- ❑ An activation function takes in weighted data (matrix multiplication between input data and weights) and outputs a non-linear transformation of the data.
- ❑ Activation functions expand our capacity to capture non-linear relationships between inputs and outputs.
- ❑ Sigmoid is an activation function, an activation function tells if a neuron is firing or not firing

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

ACTIVATION FUNCTIONS

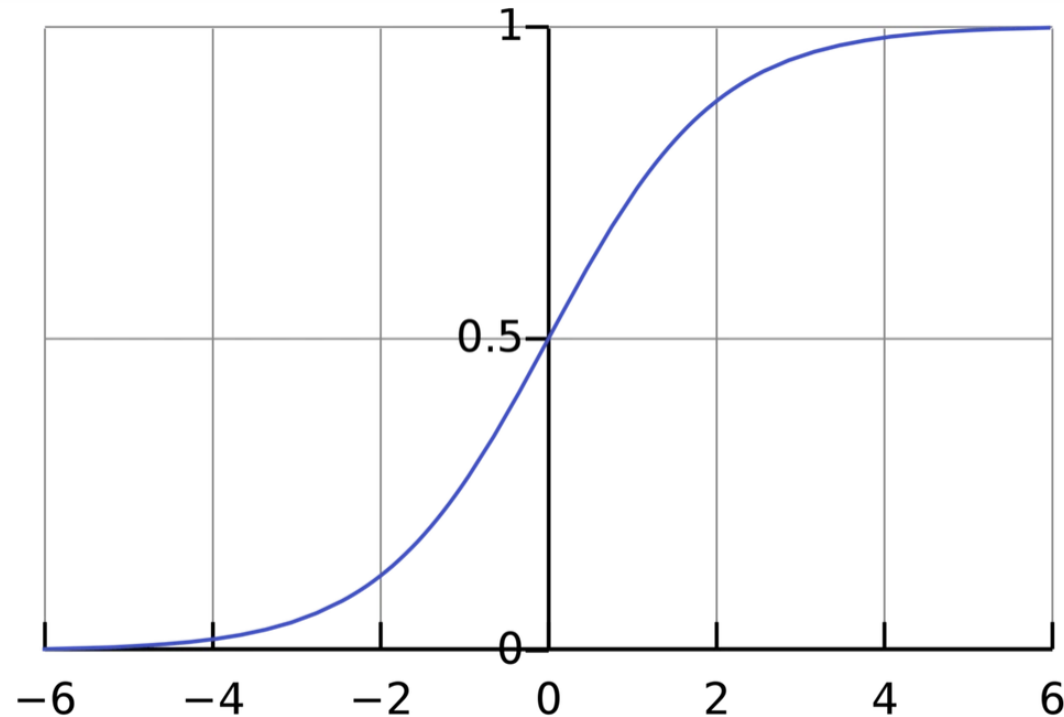
- We can see that $\sigma(z)$ acts as a sort of “**squashing**” function, condensing our previously unbounded output to the range 0 to 1.

$$z = b + \sum_i w_i x_i$$

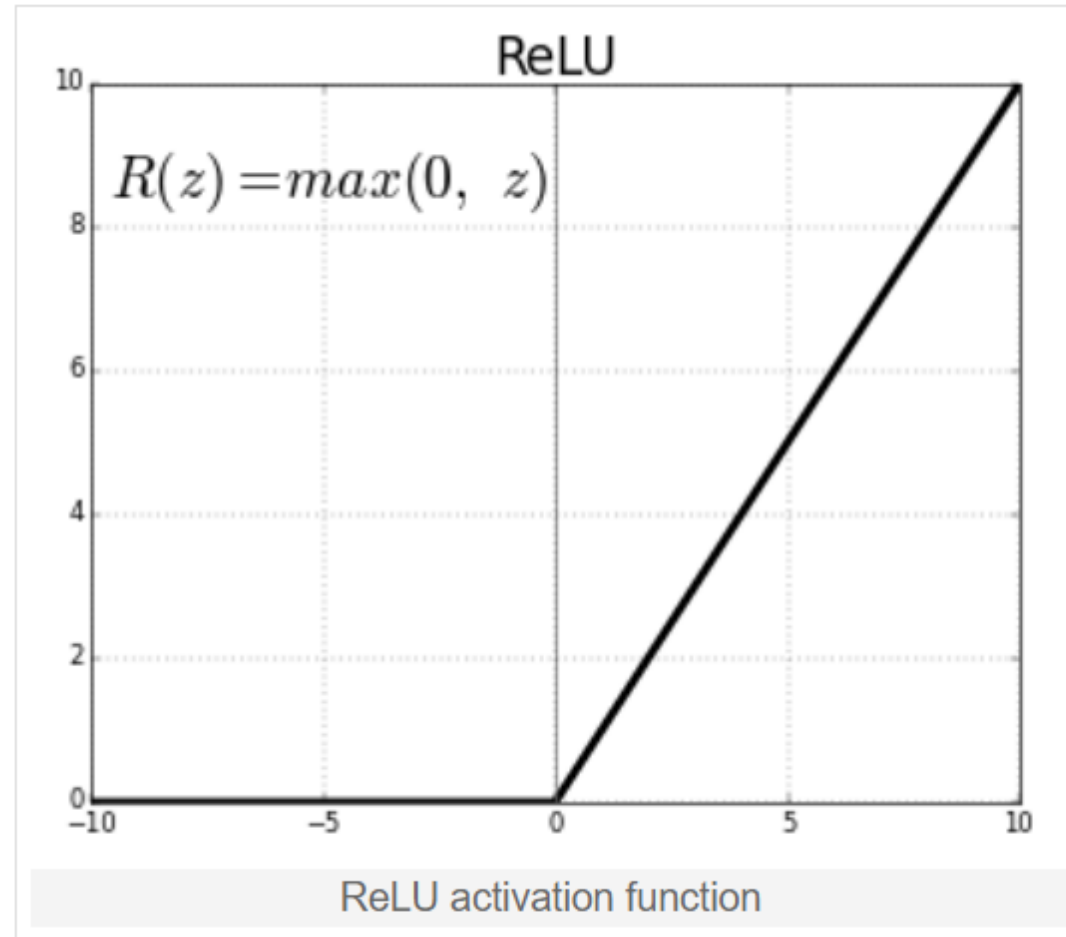
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

SIGMOID FUNCTION

The activation function turns an unbounded input into an output with a nice, predictable form. A commonly used activation function is the

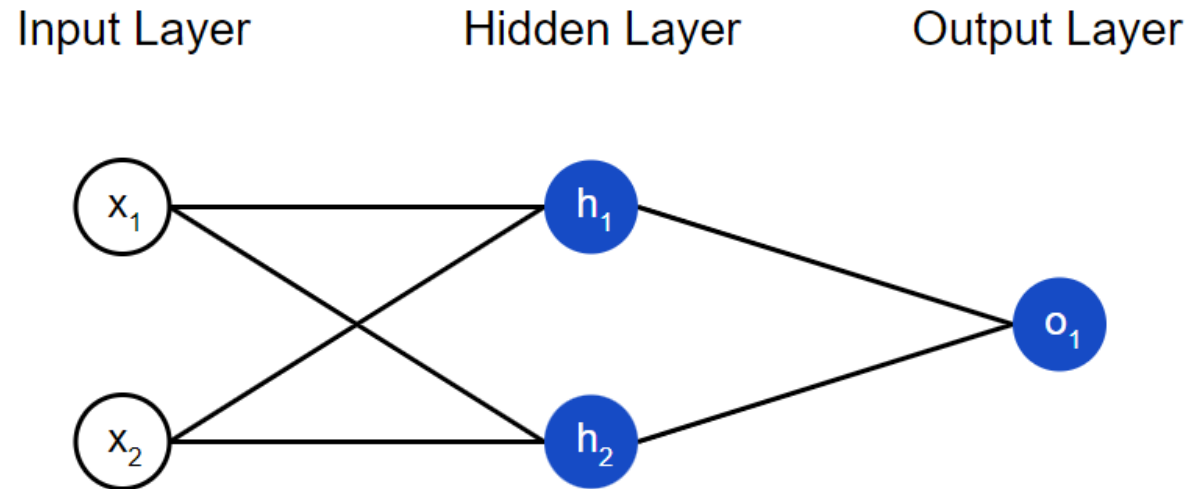


RELU ACTIVATION FUNCTION



NEURAL NETWORK

A neural network is nothing more than a bunch of neurons connected together. Here's what a simple neural network might look like:



NEURAL NETWORK WORKINGS

- Let's train our network to predict someone's gender given their weight and height:

Name	Weight (lb)	Height (in)	Gender
Alice	133	65	F
Bob	160	72	M
Charlie	152	70	M
Diana	120	60	F

- We'll represent Male with a 0 and Female with a 1, and we'll also shift the data to make it easier to use:

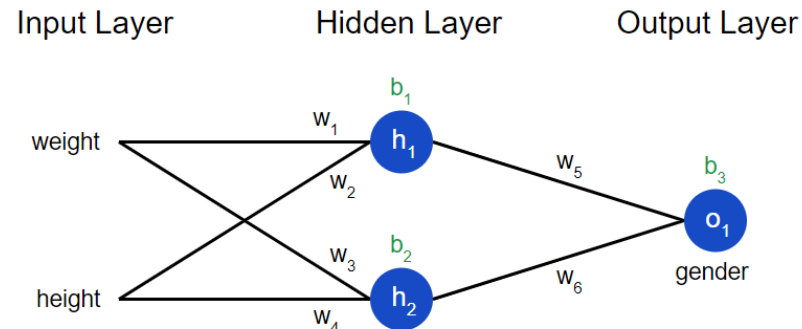
Name	Weight (minus 135)	Height (minus 66)	Gender
Alice	-2	-1	1
Bob	25	6	0
Charlie	17	4	0
Diana	-15	-6	1

NEURAL NETWORK WORKINGS

- We'll use the **mean squared error** (MSE) loss to quantify how good the model is:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_{\text{true}} - y_{\text{pred}})^2$$

- Training a network = trying to minimize its loss.
- Another way to think about loss is as a function of weights and biases.



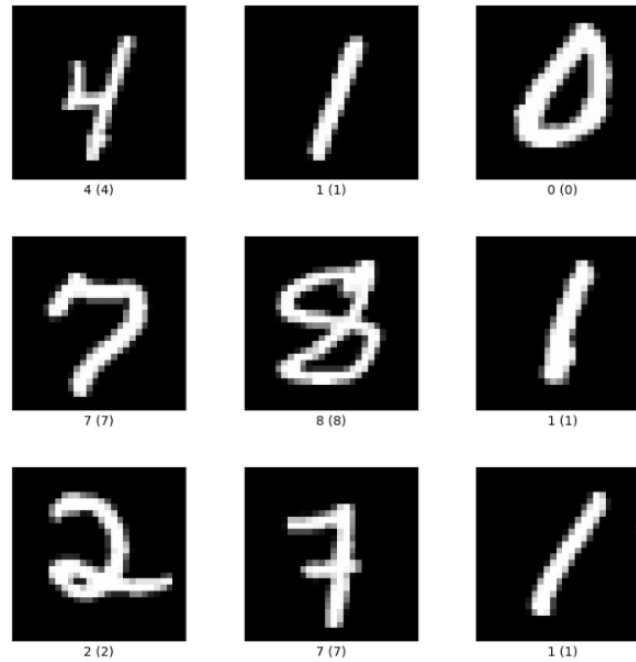
TENSORFLOW

- ❑ Tensorflow is an open-source library for Numerical computation and large scale machine learning
- ❑ Tensors are the standard way of representing data
- ❑ It includes a feature that defines, optimizes and calculates mathematical expressions easily with the help of multi-dimensional arrays called tensors.
- ❑ It includes programming support of deep neural networks and machine learning techniques.
- ❑ It includes a high scalable feature of computation with various data sets.
- ❑ TensorFlow uses GPU computing, automating management. It also includes a unique feature of optimization of the same memory and the data used.

KERAS

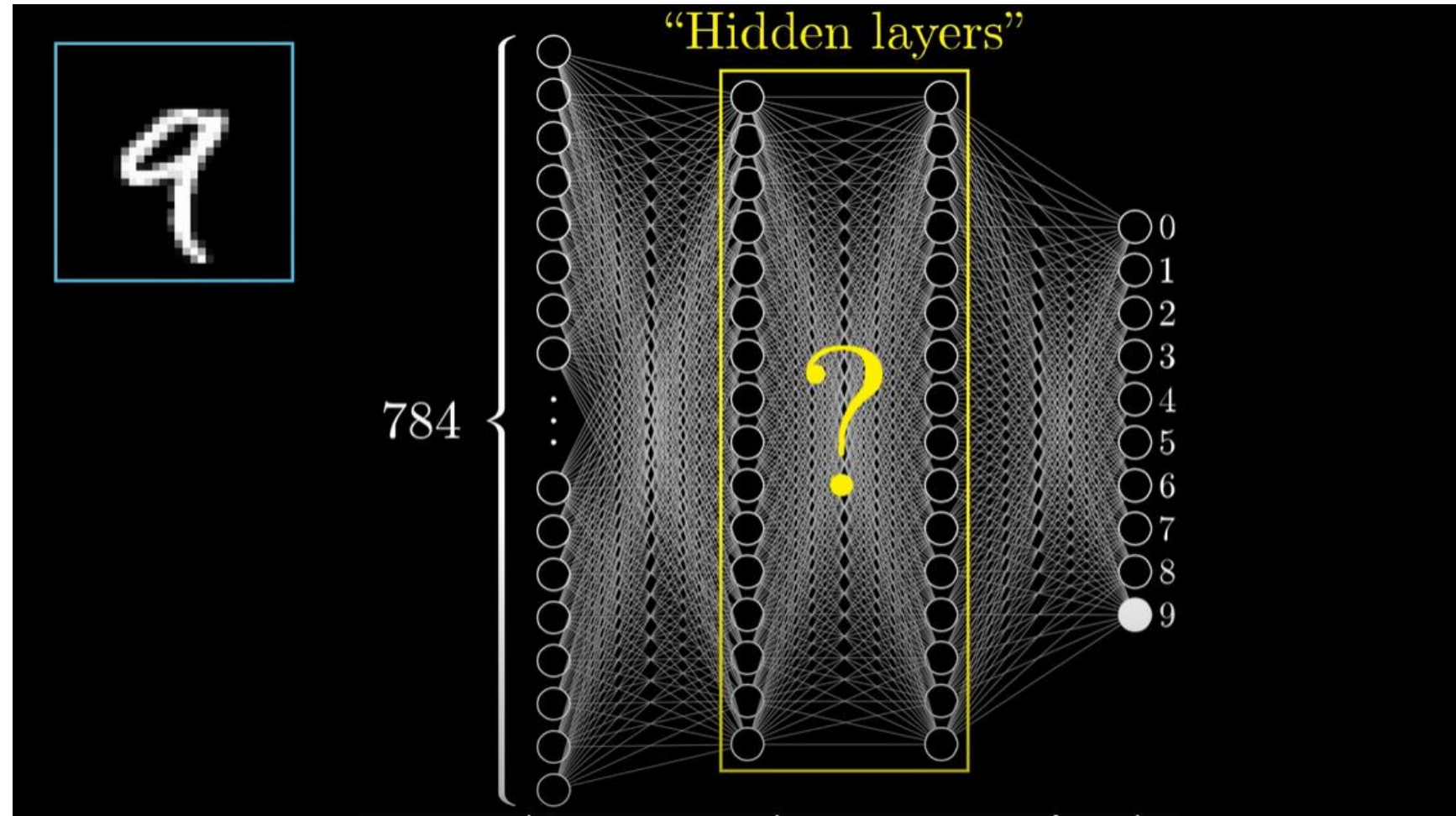
- ❑ Keras is a high-level, deep learning API developed by Google for implementing neural networks.
- ❑ Keras is an Open Source Neural Network library written in Python that runs on top of Theano or Tensorflow.
- ❑ Keras is relatively easy to learn and work with because it provides a python frontend with a high level of abstraction while having the option of multiple back-ends for computation purposes. This makes Keras slower than other deep learning frameworks, but extremely beginner-friendly.
- ❑ TensorFlow has adopted Keras as its official high-level API. Keras is embedded in TensorFlow and can be used to perform deep learning fast as it provides inbuilt modules for all neural network computations.

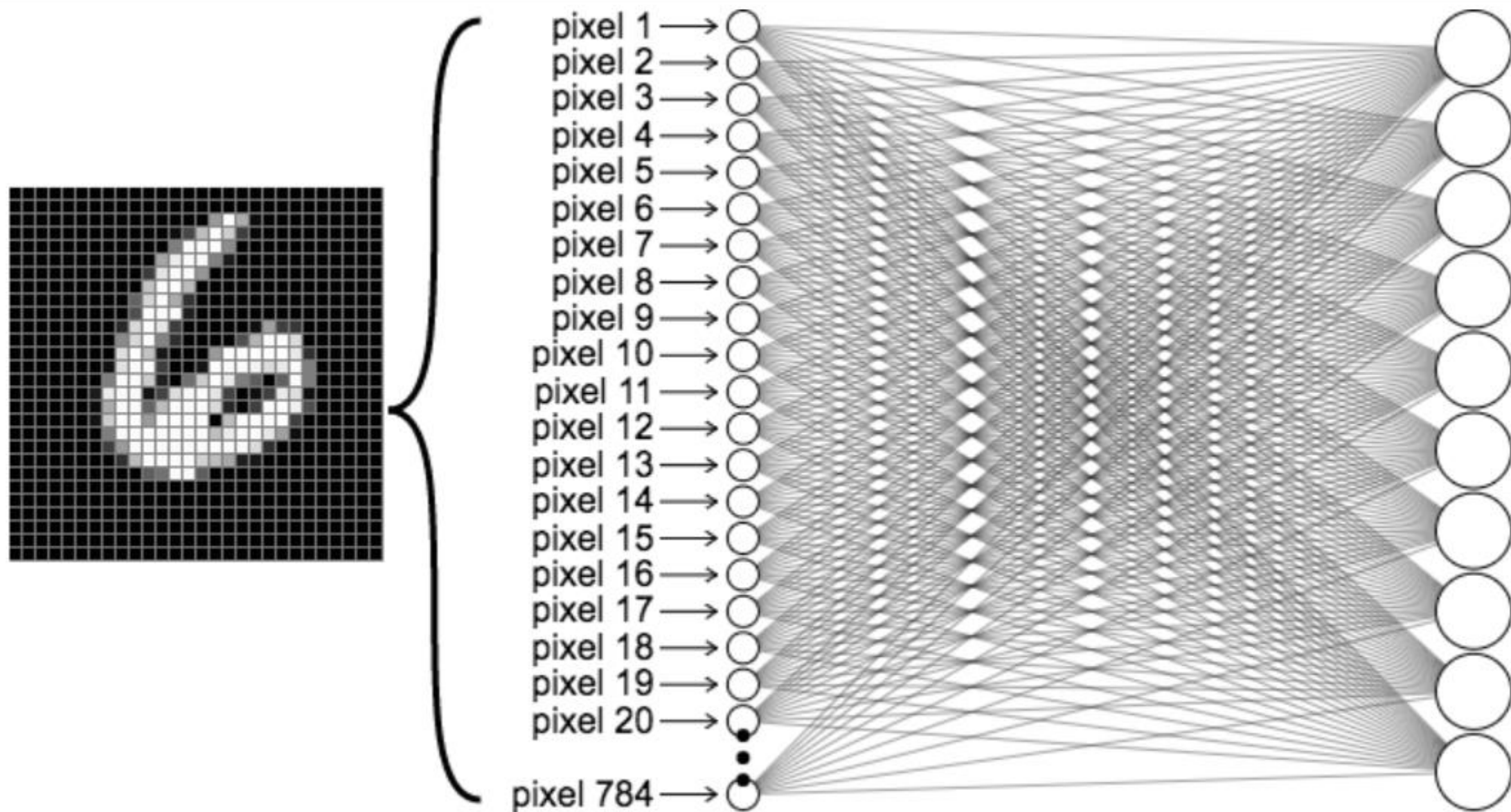
NEURAL NETWORK FOR DIGIT CLASSIFICATION



WHAT IS NEURAL NETWORK – AN INTUITIVE WAY

- ❑ Neural Networks are inspired by the brain.
- ❑ Neuron: is a function

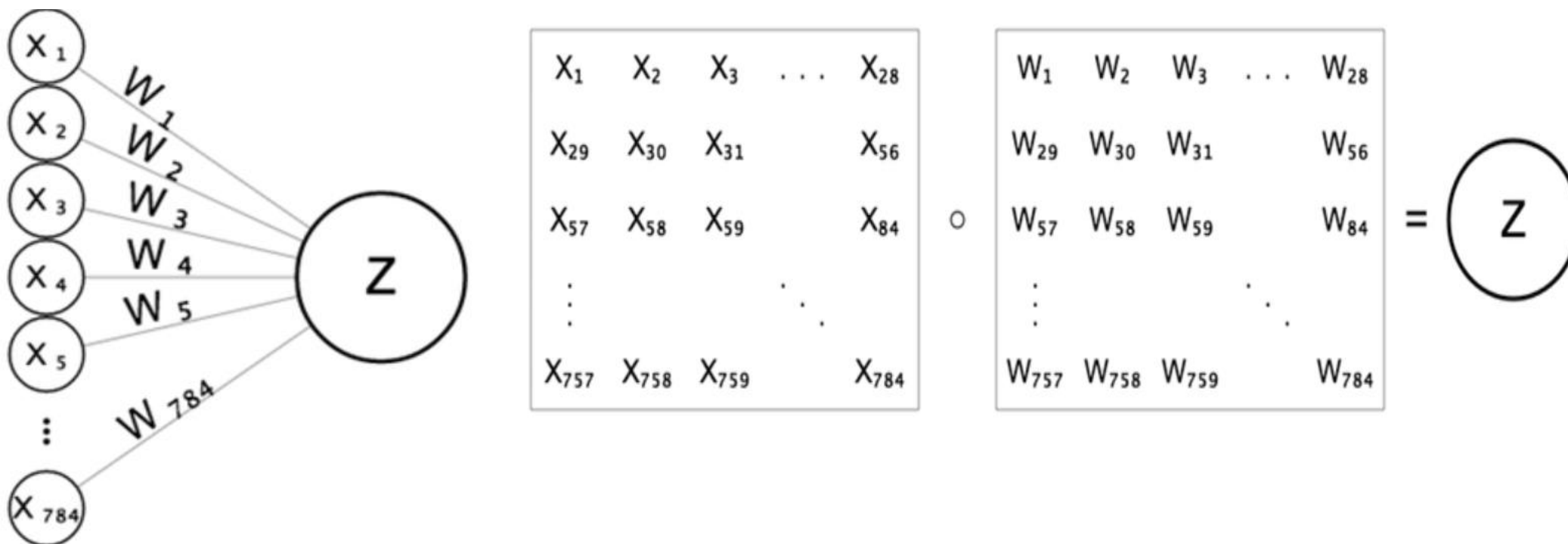




1-layer neural network for MNIST. The 10 output neurons correspond to our classes, the 10 digits from 0 to 9.

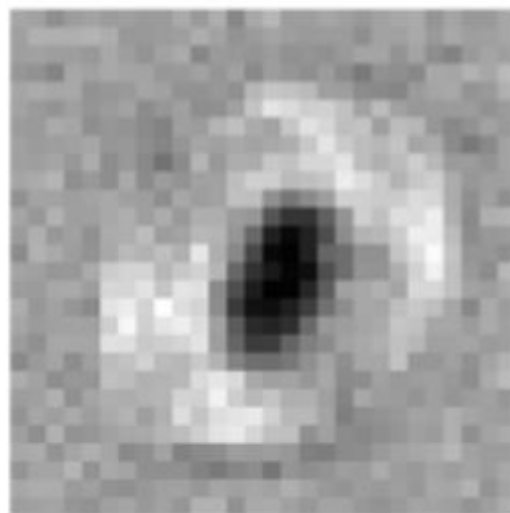
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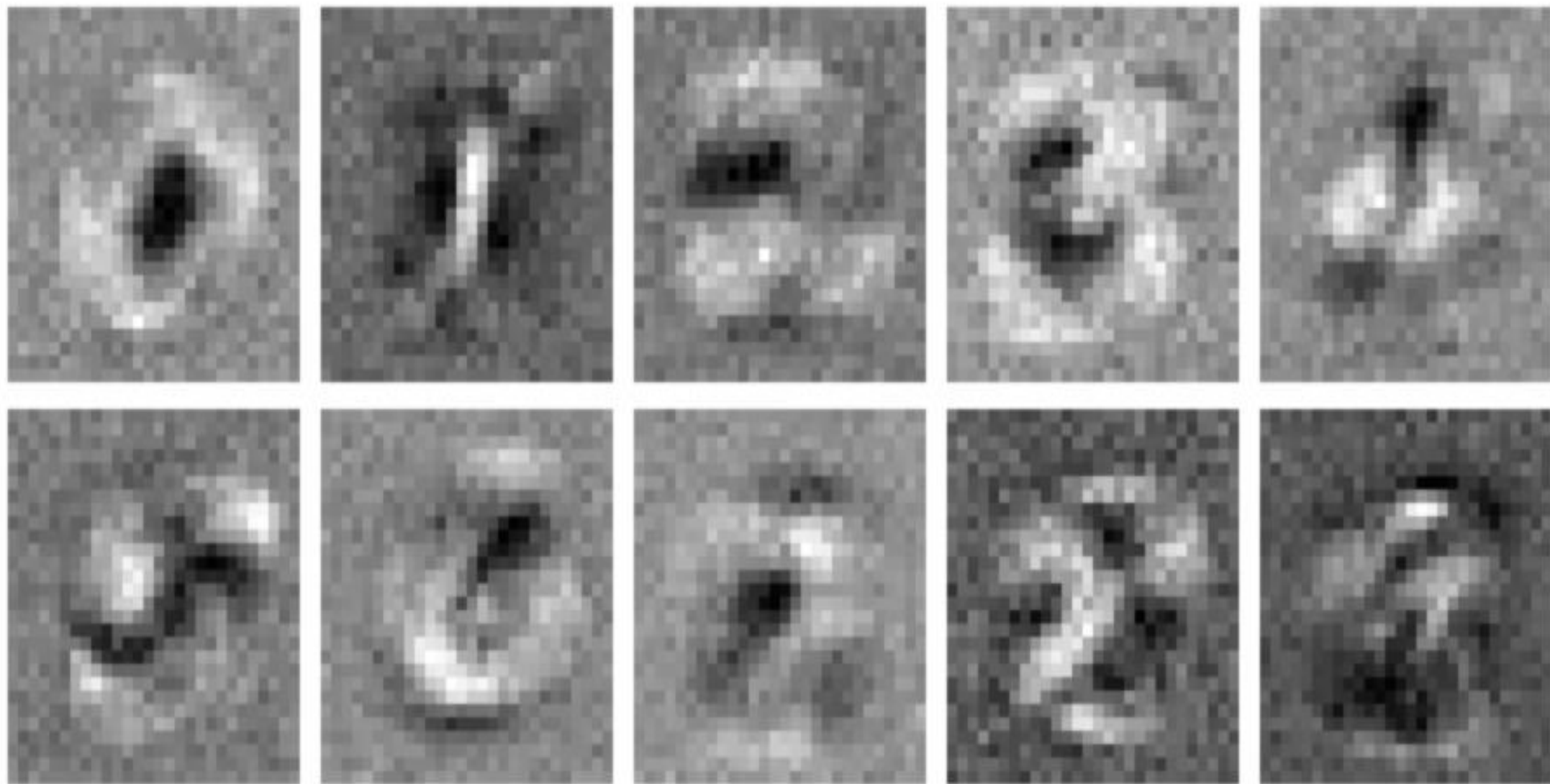




$$z = b + \sum wx$$

W_1	W_2	W_3	$\dots$	W_{28}
W_{29}	W_{30}	W_{31}		W_{56}
W_{57}	W_{58}	W_{59}		W_{84}
$\vdots$			$\ddots$	
W_{757}	W_{758}	W_{759}		W_{784}





Visualizing the weights for all the output neurons of an MNIST classifier

